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This hands-on cybersecurity project simulates a controlled authentication attack against a Windows Server 2022 system and implements an end-to-end detection and alerting workflow using Splunk Enterprise SIEM.

The objective is to gain practical experience with Windows security telemetry, SPL query development, authentication monitoring, and SOC detection and alerting.

Detection Pipeline

Kali Linux

↓

Controlled Authentication Attempts

↓

Windows Server 2022

↓

Windows Security Event Logs

↓

Splunk Universal Forwarder

↓

Splunk Enterprise SIEM

↓

SPL Detection & Aggregation

↓

Automated SOC Alert

Target Account Preparation

A dedicated test account (testuser) was created on the Windows Server to provide a controlled target for authentication testing without using an administrative account.

```
PS C:\> net user testuser * /add
Type a password for the user:
Retype the password to confirm:
The command completed successfully.

PS C:\> net user testuser
User name                testuser
Full Name
Comment
User's comment
Country/region code      000 (System Default)
Account active           Yes
Account expires          Never

Password last set        9/10/2026 1:58:05 PM
Password expires         10/22/2026 1:58:05 PM
Password changeable      9/11/2026 1:58:05 PM
Password required        Yes
User may change password Yes

Workstations allowed     All
Logon script
User profile
Home directory
Last logon              Never

Logon hours allowed      All

Local Group Memberships
Global Group memberships *Domain Users
The command completed successfully.

PS C:\>
```

Controlled Failed Authentication Attempts

Connectivity to the Windows Server (192.168.50.40) was verified from the Kali attacker machine. A targeted Nmap scan confirmed that **SMB (445)** and **RDP (3389)** were accessible, identifying suitable services for controlled authentication testing.

```
joanna@kali: ~
File Actions Edit View Help

(joanna@kali)-[~]
$ ping -c 3 192.168.50.40
PING 192.168.50.40 (192.168.50.40) 56(84) bytes of data:
64 bytes from 192.168.50.40: icmp_seq=1 ttl=128 time=0.494 ms
64 bytes from 192.168.50.40: icmp_seq=2 ttl=128 time=0.472 ms
64 bytes from 192.168.50.40: icmp_seq=3 ttl=128 time=0.475 ms

--- 192.168.50.40 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2028ms
rtt min/avg/max/mdev = 0.472/0.480/0.494/0.009 ms

(joanna@kali)-[~]
$ nmap -Pn -p 3389,445 192.168.50.40
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-09-14 19:38 BST
Nmap scan report for 192.168.50.40
Host is up (0.00053s latency).

PORT      STATE SERVICE
445/tcp   open  microsoft-ds
3389/tcp   open  ms-wbt-server

Nmap done: 1 IP address (1 host up) scanned in 0.55 seconds

(joanna@kali)-[~]
$
```

A controlled RDP login attempt was performed from the Kali attacker machine against the testuser account on the Windows Server. An intentionally incorrect password resulted in STATUS_LOGON_FAILURE, confirming that failed remote authentication activity was successfully generated.

```
joanna@kali: ~  
File Actions Edit View Help  
joanna@kali)-[~]  
$ xfreerdp /v:192.168.50.40 /u:testuser  
[19:42:52:953] [9256:9265] [INFO][com.freerdp.crypto] - creating directory /home/joanna/.config/freerdp  
[19:42:52:953] [9256:9265] [INFO][com.freerdp.crypto] - creating directory [/home/joanna/.config/freerdp/certs]  
[19:42:52:953] [9256:9265] [INFO][com.freerdp.crypto] - created directory [/home/joanna/.config/freerdp/server]  
[19:42:55:043] [9256:9265] [WARN][com.freerdp.crypto] - Certificate verification failure 'self-signed certificate (18)' at stack position 0  
[19:42:55:043] [9256:9265] [WARN][com.freerdp.crypto] - CN = Win-Serv-22.siwia.local  
[19:42:55:044] [9256:9265] [ERROR][com.freerdp.crypto] - @  
[19:42:55:044] [9256:9265] [ERROR][com.freerdp.crypto] - @ WARNING: CERTIFICATE NAME MISMATCH!  
[19:42:55:045] [9256:9265] [ERROR][com.freerdp.crypto] - @  
[19:42:55:045] [9256:9265] [ERROR][com.freerdp.crypto] - The hostname used for this connection (192.168.50.40:3389)  
[19:42:55:045] [9256:9265] [ERROR][com.freerdp.crypto] - does not match the name given in the certificate:  
[19:42:55:045] [9256:9265] [ERROR][com.freerdp.crypto] - Common Name (CN):  
[19:42:55:045] [9256:9265] [ERROR][com.freerdp.crypto] - Win-Serv-22.siwia.local  
[19:42:55:045] [9256:9265] [ERROR][com.freerdp.crypto] - A valid certificate for the wrong name should NOT be trusted!  
Certificate details for 192.168.50.40:3389 (RDP-Server):  
Common Name: Win-Serv-22.siwia.local  
Subject: CN = Win-Serv-22.siwia.local  
Issuer: CN = Win-Serv-22.siwia.local  
Thumbprint: 78:40:9e:a1:98:22:7c:6d:4c:37:6f:93:84:19:ed:5c:c5:f5:89:0f:cb:6d:30:de:8a:b0:99:70:a9:89:ab:7f  
The above X.509 certificate could not be verified, possibly because you do not have the CA certificate in your certificate store, or the certificate has expired.  
Please look at the OpenSSL documentation on how to add a private CA to the store.  
Do you trust the above certificate? (Y/T/N) y  
Password:  
[19:44:04:157] [9256:9265] [WARN][com.freerdp.core.nla] - SPNEGO received NTSTATUS: STATUS_LOGON_FAILURE [0xC000006D] from server  
[19:44:04:157] [9256:9265] [ERROR][com.freerdp.core] - nla_recv_pdu:freerdp_set_last_error ex ERRCONNECT_LOGON_FAILURE [0x00020014]  
[19:44:04:157] [9256:9265] [ERROR][com.freerdp.core.rdp] - rdp_recv_callback: CONNE
```

Controlled RDP Authentication Attack

RDP (3389) was selected as the authentication vector. THC Hydra was used from Kali Linux to generate multiple failed login attempts against the dedicated testuser account. A small custom password list was used to generate repeated authentication failures.

```
hydra -l testuser -P pass.txt rdp://192.168.50.40
```

The attack completed with no valid password found, generating failed authentication telemetry for analysis in Splunk.

```
(joanna@kali)-[~]
$ printf "password\njoanna\nAdmin\ntest\nEasyPassword\nSummer\n" > pass.txt

(joanna@kali)-[~]
$ hydra -l testuser -P pass.txt rdp://192.168.50.40
Hydra v9.5 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service
organizations, or for illegal purposes (this is non-binding, these *** ignore laws and ethics anyway)

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-09-14 19:52:11
[WARNING] rdp servers often don't like many connections, use -t 1 or -t 4 to reduce the number of parallel
connections and -W 1 or -W 3 to wait between connection to allow the server to recover
[INFO] Reduced number of tasks to 4 (rdp does not like many parallel connections)
[WARNING] the rdp module is experimental. Please test, report - and if possible, fix.
[DATA] max 4 tasks per 1 server, overall 4 tasks, 6 login tries (l:1/p:6), ~2 tries per task
[DATA] attacking rdp://192.168.50.40:3389/
1 of 1 target completed, 0 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-09-14 19:52:14

(joanna@kali)-[~]
$
```

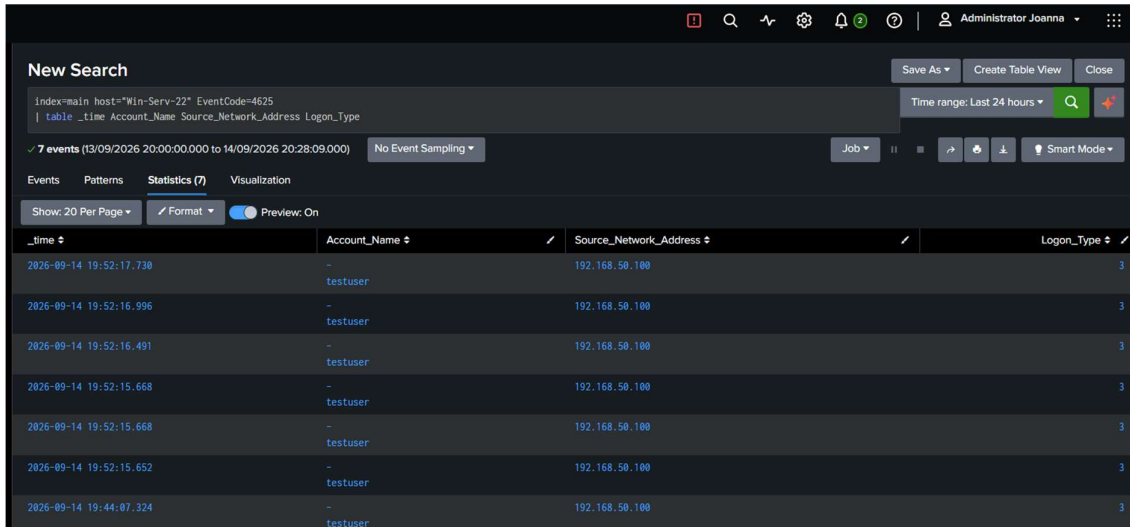
Splunk Investigation

Splunk was used to identify Windows EventCode 4625, representing failed logon attempts. The events identified repeated authentication failures against testuser originating from the Kali attacker (192.168.50.100).

index=main host="Win-Serv-22" EventCode=4625

| table _time Account_Name Source_Network_Address Logon_Type

The results successfully correlate the simulated RDP authentication attack with Windows Security telemetry.



The screenshot shows the Splunk interface with a search query: `index=main host="Win-Serv-22" EventCode=4625 | table _time Account_Name Source_Network_Address Logon_Type`. The results are displayed in a table with 7 events. The columns are `_time`, `Account_Name`, `Source_Network_Address`, and `Logon_Type`. All events show failed logon attempts for the `testuser` from the IP address `192.168.50.100`.

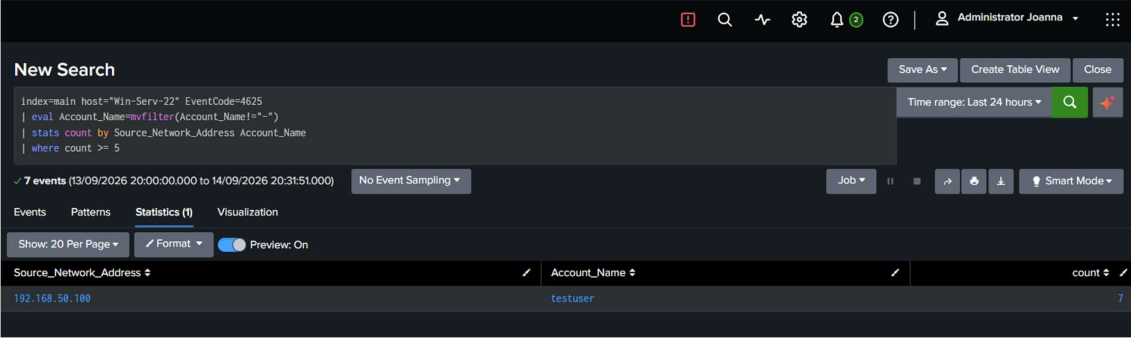
_time	Account_Name	Source_Network_Address	Logon_Type
2026-09-14 19:52:17.730	testuser	192.168.50.100	3
2026-09-14 19:52:16.996	testuser	192.168.50.100	3
2026-09-14 19:52:16.491	testuser	192.168.50.100	3
2026-09-14 19:52:15.668	testuser	192.168.50.100	3
2026-09-14 19:52:15.668	testuser	192.168.50.100	3
2026-09-14 19:52:15.652	testuser	192.168.50.100	3
2026-09-14 19:44:07.324	testuser	192.168.50.100	3

Detection Engineering

A Splunk detection query was created to identify multiple failed authentication attempts originating from the same source. The detection identified seven failed logons against the testuser account from the Kali attacker at 192.168.50.100.

Detection query:

```
index=main host="Win-Serv-22" EventCode=4625  
| eval Account_Name=mvfilter(Account_Name!="-")  
| stats count by Source_Network_Address Account_Name  
| where count >= 5
```



The screenshot shows the Splunk Search interface. At the top, the 'New Search' panel displays the query: `index=main host="Win-Serv-22" EventCode=4625 | eval Account_Name=mvfilter(Account_Name!="-") | stats count by Source_Network_Address Account_Name | where count >= 5`. The search results show 7 events for the time range 13/09/2026 20:00:00.000 to 14/09/2026 20:31:51.000. The results table has three columns: Source_Network_Address, Account_Name, and count. The data row shows the IP address 192.168.50.100, the account name testuser, and a count of 7.

Source_Network_Address	Account_Name	count
192.168.50.100	testuser	7

A scheduled Splunk alert was created using the detection query to identify five or more failed logon attempts from the same source IP. The alert was configured to run every five minutes with High severity.

Summary

This scenario demonstrated the detection of an RDP brute-force attack against a Windows Server. Multiple failed authentication attempts generated Windows Security Event ID 4625, which were forwarded to Splunk and correlated with the Kali attacker IP. A detection rule was then created to identify repeated failed logons and configured as a scheduled High-severity alert. The final test successfully triggered the alert, confirming that the detection and monitoring workflow operated as intended.